Customer Intelligence Platform — Churn & Lifetime Value Analytics

End-to-End Master Project Documentation & Interview Guide

Author: Data Analyst & ML Engineer	**Tech Stack:** Python, SQL, Scikit-learn, Power BI
Scope: 10,000 Customers | 60,597 Transactions	**Target Metrics:** 98.19% ROC-AUC | 0.9536 R^2

1. Executive Summary & Core Business Problem

In modern retail and e-commerce, retaining existing customers is 5x cheaper than acquiring new ones. Traditional business dashboards provide **hindsight** (reporting past sales), whereas this platform delivers **foresight** by predicting customer churn 90 days before drop-off occurs and forecasting 12-month Customer Lifetime Value (CLV).

2. Step-by-Step Task Breakdown (Who Did What)

To ensure full transparency during technical interviews, the table below outlines the division of work between the **Agent (Antigravity AI)** and **You (The User / Engineer)** across every milestone:

Milestone	Agent Accomplishments (Backend)	Your Accomplishments (Execution & BI)
1. Environment & Setup	Created workspace structure (`src/`, `sql/`, `docs/`, `data/`) & dependency lists.	Installed Python libraries (`pandas`, `scikit-learn`, `powerbi`) in terminal.
2. Data Engineering	Wrote `src/generate_dataset.py` using Beta, Lognormal & Poisson distributions.	Executed dataset script generating 10,000 profiles & 60,597 transaction logs.
3. RFM & Clustering	Wrote ANSI SQL query (`sql/rfm_analysis.sql`) & K-Means clustering script (`src/rfm_segmentation.py`).	Executed segmentation script, validated Silhouette Score (k=5), assigned RFM personas.
4. Churn ML Classifier	Wrote Scikit-learn classification pipeline (`src/churn_prediction.py`) comparing Logistic Regression vs Random Forest.	Executed training, achieved **98.19% ROC-AUC**, generated customer risk tiers.
5. CLV ML Regressor	Wrote Scikit-learn regression pipeline (`src/clv_estimation.py`) comparing Random Forest Regressor vs Ridge.	Executed regressor, achieved **0.9536 R^2 Score ($29.37 MAE)**, assigned VIP value tiers.
6. Power BI Dashboard	Designed Star Schema data model & wrote DAX measures (`Total Revenue`, `Churn %`, `Avg CLV`).	Loaded CSVs in Power BI Desktop, established 1:* relationship, constructed 3-page interactive visual dashboard.

3. Algorithm Selection & Technical Justifications

A. Unsupervised Learning: K-Means Clustering (`src/rfm_segmentation.py`)

Why we used it: To group customers into natural behavioral clusters without manual biases.

How it works: Standardized RFM features (`StandardScaler`) and minimized within-cluster sum-of-squares (WCSS). Evaluated silhouette scores across k=3..6 to select optimal **k=5 clusters** (Silhouette Score: `0.2236`).

Customer Personas Created: Champions (18.6%), Loyal Customers (20.2%), At-Risk Spenders (12.8%), Hibernating (23.9%), Lost (20.3%).

B. Supervised Classification: Churn Risk Prediction (`src/churn_prediction.py`)

Why we used it: To estimate individual customer churn probability (0.00 to 1.00) 90 days before drop-off.

Algorithms Compared: Logistic Regression, Random Forest Classifier, Gradient Boosting.

Winning Model: Logistic Regression / Random Forest

Performance Metrics: ROC-AUC = **98.19%** | F1-Score = **0.9107** | Precision = **96.93%** | Recall = **85.88%**

Core Business Finding: Customers opening **3+ support tickets** paired with shipping delays > 3 days experienced a churn probability spike past **75%**.

C. Supervised Regression: 12-Month CLV Estimation (`src/clv_estimation.py`)

Why we used it: To forecast future 12-month expected dollar revenue per customer.

Algorithms Compared: Random Forest Regressor, Ridge Regression, Gradient Boosting Regressor.

Winning Model: Random Forest Regressor (`n_estimators=150, max_depth=10`)

Performance Metrics: R^2 Score = **0.9536** (explains 95.4% revenue variance) | MAE = **$29.37**

Value Tiers: Tier 1 VIP ($780+), Tier 2 High ($260-$780), Tier 3 Medium ($120-$260), Tier 4 Low (<$120).

4. File-by-File Technical Blueprint

Filename & Path	Technical Purpose	Output Asset Produced
`Customer_Intelligence_Dashboard.pbix`	Interactive 3-page Power BI report.	Visual executive dashboard
`churn_clv_rfm_analytics.ipynb`	Executed Jupyter analytics notebook with charts.	EDA & model validation plots
`src/generate_dataset.py`	Synthetic data engine using Beta/Lognormal math.	`customer_profiles.csv` & `transactions.csv`
`src/rfm_segmentation.py`	Quantile scoring & K-Means clustering script.	`rfm_segmented_customers.csv`
`src/churn_prediction.py`	Scikit-learn classification pipeline.	`churn_predictions.csv` (ROC-AUC 98.2%)
`src/clv_estimation.py`	Scikit-learn regression pipeline.	`clv_predictions.csv` (R^2 0.954)
`sql/rfm_analysis.sql`	ANSI SQL queries using CTEs & `NTILE(5)`.	SQL feature extraction logic
`docs/power_bi_dashboard_spec.md`	DAX measures & Star Schema layout guide.	Power BI architectural blueprint
`README.md`	GitHub repository documentation & architecture.	Portfolio front page documentation

5. Power BI Star Schema & DAX Measures

Star Schema Model: Connected `Dim_Customer_Analytics` (1) ■■■■ (*) `Fact_Transactions` on `customer_id` with single-direction filtering.

```
Total Revenue = SUM(Fact_Transactions[order_amount])
Total Active Customers = CALCULATE(COUNTROWS(Dim_Customer_Analytics),
Dim_Customer_Analytics[is_churned] = 0)
Churn Rate % = DIVIDE(CALCULATE(COUNTROWS(Dim_Customer_Analytics), Dim_Customer_Analytics[is_churned]
= 1), COUNTROWS(Dim_Customer_Analytics), 0)
Avg Predicted 12M CLV = AVERAGE(Dim_Customer_Analytics[predicted_12m_clv])
At Risk Revenue = CALCULATE(SUM(Dim_Customer_Analytics[monetary]), Dim_Customer_Analytics[risk_tier] =
"High Risk")
```

6. Top Technical Interview Q&A; Playbook

Q1: How did you prevent data leakage during machine learning model training?

Answer: I enforced strict featurization ordering by splitting the raw dataset into 80% Train and 20% Test sets BEFORE fitting any preprocessors like StandardScaler or OneHotEncoder. I fit transformers exclusively on training data and transformed test data independently.

Q2: Why evaluate ROC-AUC and F1-Score instead of just Accuracy?

Answer: Accuracy is misleading in imbalanced datasets. ROC-AUC evaluates the model's capacity to rank high-risk customers above low-risk customers across all thresholds, while F1-Score balances Precision (avoiding false alarms) and

Recall (catching actual churners). My model achieved 98.19% ROC-AUC.

Q3: How did you select k=5 in K-Means Clustering?

Answer: I standardized RFM features using StandardScaler and evaluated cluster separation across k=3 to 6 using Silhouette Scores. The highest silhouette score of 0.2236 occurred at k=5, cleanly partitioning customers into Champions, Loyal, Potential Loyalists, At Risk, and Lost segments.

Q4: What operational insight did your model discover for executive decision making?

Answer: Feature importances proved that opening 3+ support tickets paired with shipping delays > 3 days caused customer churn probability to spike past 75%. In Power BI, I created an 'At-Risk Revenue' card ($1.30M exposure) and a live High-Risk VIP action table for immediate CSM retention calls.